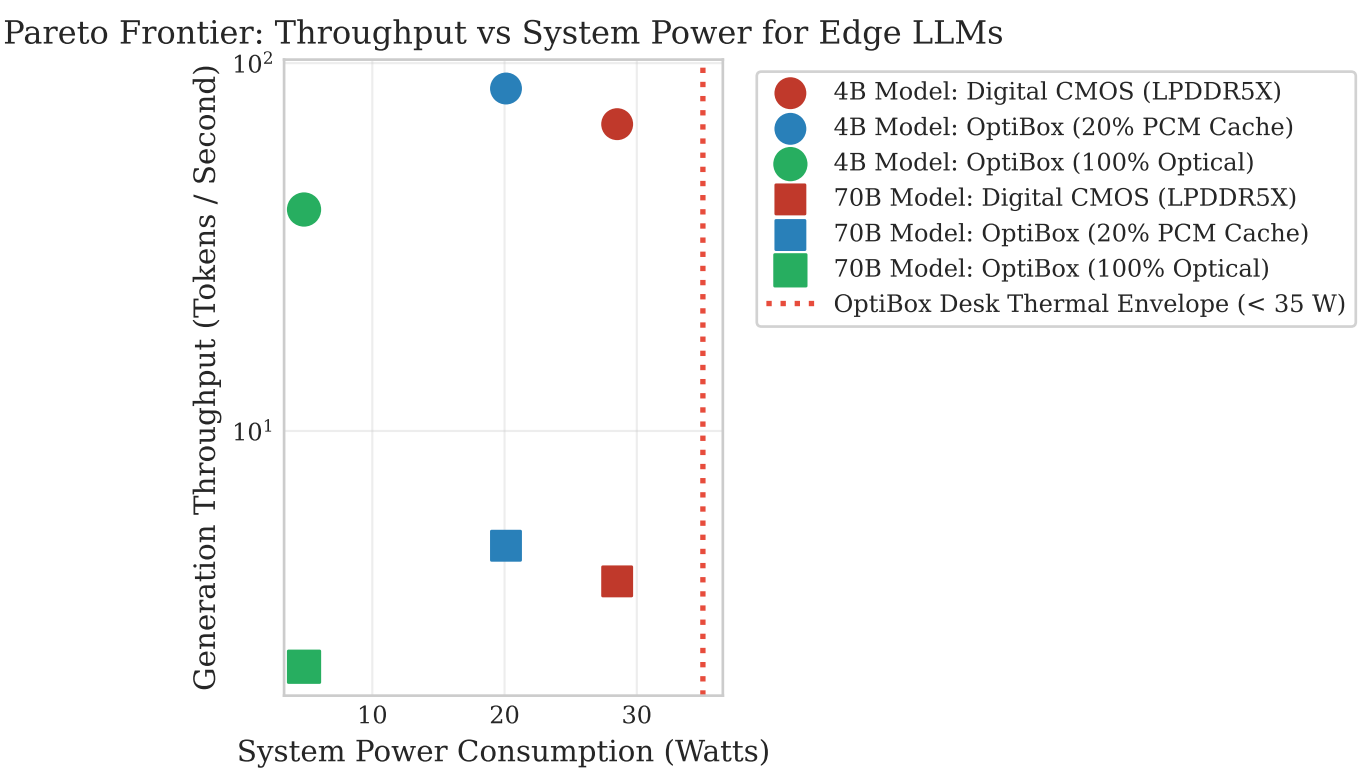
Pareto Frontier: Throughput vs System Power for Edge LLMs
Generation Throughput (Tokens / Second)
10^2
10^1
10
20
30
System Power Consumption (Watts)
4B Model: Digital CMOS (LPDDR5X)
4B Model: OptiBox (20% PCM Cache)
4B Model: OptiBox (100% Optical)
70B Model: Digital CMOS (LPDDR5X)
70B Model: OptiBox (20% PCM Cache)
70B Model: OptiBox (100% Optical)
OptiBox Desk Thermal Envelope (< 35 W)